Gates before money.

Phases, the raise, early hires and how it earns.

← all documents · Kat Titterton · San Francisco

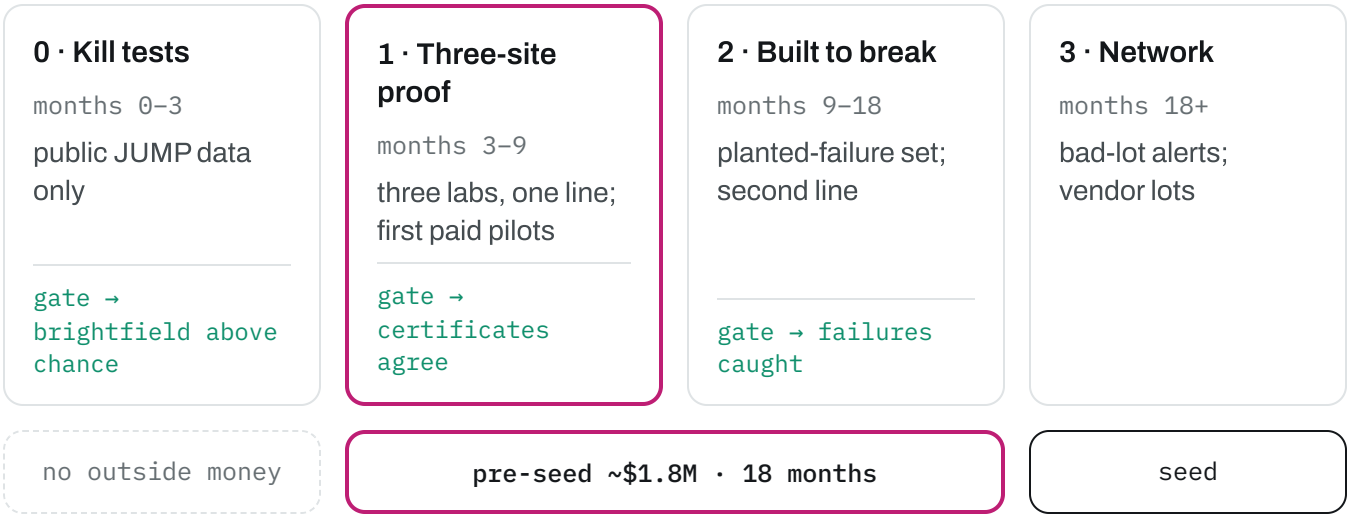
Summary

The cheapest tests are also the ones that could kill the company, so they run first, on public data, before any money is raised. Then raise about **$1.8M for 18 months**, the deck's pre-seed, to a first paid certificate: a leader, an ML engineer, the benchmark playground and three partner labs, every line planned at twice its cost. The seed round after that is raised on proof that labs upload and certificates agree, not on the science alone.

All dollar figures here are proposals derived from public benchmarks and salary norms, not quotes; market medians are cited where used.

01 Phases and gates

Fig. 1 Each phase opens only when the test before it passes



Money follows gates, never the reverse. Gates are the falsifiable tests in the white paper, §5.

Phase 0 costs a salary and compute, and it is the phase most likely to change the plan: if brightfield alone cannot place a lab, the certificate needs one cheap stain; e8 and e10 now show that answers move with position, so the pitch is not QC alone.

PHASE	MONTHS	PROVES	KILLS OR NARROWS IT
0 · Kill tests	0–3	brightfield carries the lab signal; position changes the drug answer (e8); the signal is not only instrument	brightfield near 9% chance; e8 Spearman near zero; variance nearly all microscope model
1 · Three-site proof	3–9	three labs upload the same line and their certificates agree with independent checks	labs will not upload; certificates disagree with ground truth
2 · Built to break	9–18	the planted-failure reference catches each failure and grades correction methods	failures go undetected; a second line does not transfer
3 · Network	18+	many labs per lot make bad-lot alerts possible	too few labs share a lot

02 The raise

Recommendation: about $1.8M, the deck's pre-seed, to fund 18 months to a first paid certificate. The kill tests run first on public data; the round buys the benchmark playground, three partner labs and a model for any image, then the first paid certificate.

Why that size: the $1–2.5M middle of the pre-seed market is thinning, down from 24% of rounds in Q1 2023 to 18% in Q1 2026 [1], and once a financing reaches about $3M, priced rounds become as common as SAFEs [2]. So $1.8M sits in the squeezed middle and is raised on what the public map already shows; about $3M is the size to choose if the kill tests land before the round. For scale, aggregators of Carta data put the mid-2026 median US priced seed at $4.1M on a $24.3M post-money, about 18% dilution [4].

OPTION	SIZE	WHEN	FOR	AGAINST
A · Pre-seed now, the deck's ask	**\~$1.8M**	**now**	**18 months to a first paid certificate**	**lands in the thinning middle [1]**
B · Pre-seed after kill tests	\~$3M	month 2–3	priced on evidence; covers phases 1–2	2–3 months of runway from savings or a grant
C · Priced seed now	\~$4M+	now	matches the seed median [4]	priced before brightfield is tested, so valued as a hypothesis

The next round is a seed of roughly the median size, raised around months 15–18 on phase 1 proof (section 6). Series A comes after the network works; the Q2 2026 median Series A was $14.7M [3].

03 Use of funds

About 71% of the money is people; the benchmark playground and the partner labs are the lines that buy the moat. Every line is planned at twice its estimated cost (estimates, not offers).

LINE	18 MONTHS	SHARE	WHAT IT BUYS
leader, to run it	$360k	20%	18 months

LINE	18 MONTHS	SHARE	WHAT IT BUYS
scientific lead (me)	$360k	20%	18 months
ML engineer	$320k	18%	from month 7
payroll overhead	$208k	12%	20% of salaries
benchmark playground	$240k	14%	~60 plates · planted faults
partner-lab bench time	$60k	3%	3 labs · controls they already image · reagent top-ups · shipping
compute and storage	$90k	5%	images · models · hosting · API
legal, IP, insurance, accounting	$90k	5%	incorporation to first contract
travel and partner visits	$40k	2%	2 benchmark partners · 3 labs
Total	**\~$1.8M**	**100%**	$884k planned × 2 for overruns

The order matters more than the split: nothing is spent on bench time until phase 0 passes, and the planted-failure set is sized by what phase 1 shows.

04 Early hires

Four hires, each staged to the phase that needs it. A leader comes before any of them, someone to run the company. The pre-seed pays for the leader and the first hire, the ML engineer from month 7; the other three come with the seed. You hold the science.

#	ROLE	STARTS	PROFILE TO LOOK FOR	WHY THEN
1	Founding ML engineer	month 7	image representation learning plus distribution shift; evaluates rigorously (grouped cross-validation, permutation nulls); fluent in the bioimage stack (CellProfiler, JUMP tooling); has shipped a benchmark or a paper someone reran	phase 0 is entirely computational
2	Assay and imaging scientist	with the seed	high-content imaging in brightfield and Cell Painting; has transferred an assay between sites; has run plates through cores and CROs; can design plates that fail on purpose	phase 1 needs three sites run the same way
3	Data platform engineer	with the seed	moves terabytes of images reliably; keeps each lab's data separate by design; watches cloud cost	uploads from outside labs start in earnest
4	Partnerships lead	with the seed (fractional first)	has sold to core facilities or pharma screening groups, or worked as a field scientist for an imaging or cell vendor; scientists trust them	converting pilots to paid lines before the seed

Where to look. ML: alumni of cell-profiling groups and of phenomics or ML-for-biology companies, and academic groups working on distribution shift. Assay: pharma screening groups, CROs, and core facility staff. Partnerships: field application scientists at imaging and cell companies.

Advisors, before hire 4. A bioimage-informatics academic from the JUMP circle (credibility and the reference); a head of screening at a pharma company (the buyer's view); a quality and standards expert who has run identity testing to published standards [5]; someone from a cell bank or vendor (the channel). Plus the employment counsel you already use, to keep the company clear of any prior-employer IP question.

Not yet. A sales team or a wet lab of your own: cores and CROs cover bench work until the planted-failure set proves what an in-house lab would add.

05 Revenue model

Identity testing already has a price: $217 per human STR sample at ATCC [5], $255 with a database search at the Johns Hopkins core [6]. That buys one identity check. The certificate is a behaviour check repeated every quarter, so it sells as a subscription per cell line per quarter; the quarter is a starting cadence, and the drift measured so far argues for more often.

BUYER	PRICE (PROPOSED)	WHAT'S AT STAKE FOR THEM	ROLE
Academic labs and cores	\~$500–1,000 per line-quarter; free for labs that contribute images	the paper	grows the network
Pharma and CRO screening groups	\~$5–15k per line-quarter, or a site license	the screen; one late-caught line cost \~$2.6M	pays for the company
Cell banks and vendors	a fee per lot certified, plus the right to ship the certificate with it	the lot and every lab that bought it	the channel
Later: bad-lot alerts	subscription	every screen run on a drifting lot	only the network can make it

Unit economics (derived, not measured). One pharma site with 10 active lines at $5k per line-quarter is \~$200k a year; twenty such sites is \~$4M a year before vendor or alert revenue. Against a single \~$2.6M failure, $50k a year is under 2% of the loss it prevents. For the size of the whole problem, about $28B a year of US preclinical research is irreproducible, with reagents and reference materials the largest share [7].

What never gets sold. Rankings of instruments, or the right for a vendor to grade its own lot. The benchmark can be licensed; the verdict cannot.

06 What the seed is raised on

The seed pitch is traction, not the 99.9%. By about month 15, these should be true:

EVIDENCE	TARGET BY MONTH 15	WHY AN INVESTOR CARES
Kill tests	all three reported, including any that narrowed the product	the risk that was science is now known
Labs uploading	3+ institutions on one line, uploading every quarter	the cold start is solved
Certificates that hold up	agreement with independent checks at all three sites	the product works outside your hands
Paying pilots	2+ paying, at least one pharma or CRO	someone pays at the proposed price
Second line	the reference extended past U2OS	the method is not one cell line
Planted-failure set	designed and started	the moat is under construction

07 Trade-offs to decide

DECISION	SHORT-TERM COST	WHAT IT BUYS	RECOMMENDATION
Free certificates for contributing labs	academic revenue	the reference and, later, bad-lot detection	✓
Price per line per quarter, not per plate	revenue that tracks usage	repeated uploads over time, which build the reference	per line
Raise after the kill tests, not before	2–3 months without outside money	a round priced on evidence	wait
Instrument makers as a channel	the easiest distribution	none that survives: neutrality is the moat	license the benchmark only

DECISION	SHORT-TERM COST	WHAT IT BUYS	RECOMMENDATION
Cores and CROs instead of an own lab	less control of the bench	18 months of runway	outsource until phase 2
Who leads it	the lead, and its equity	someone whose job is to run it, while you build the science	a leader, or a home that already has one

References

Checked for this doc on 28 September 2026. Reference 4 is a secondary summary of Carta data; the others are primary. Salaries, prices and budgets not cited here are proposals.

1. Carta. State of Pre-Seed: Q1 2026. carta.com
2. Carta. A disappearing middle: why the pre-seed market is growing increasingly barbell-shaped. March 2026. carta.com
3. Carta. Series A funding: how to raise your first priced round (Q2 2026 benchmarks). carta.com
4. Waveup. Seed round benchmarks 2026, summarising Carta medians. waveup.com
5. ATCC. FTA Sample Collection Kit for Human Cell Authentication Service (STR), 135-XV. atcc.org · service standards: atcc.org
6. Johns Hopkins Genetic Resources Core Facility. Cell line authentication. grcf.jhmi.edu
7. Freedman LP, Cockburn IM, Simcoe TS. The economics of reproducibility in preclinical research. *PLoS Biology* 13(6):e1002165, 2015. PMC